Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Lesson 1: Sum Rule, Product Rule, and Bayes' Theorem

- Sum Rule, Product Rule, and Bayes' Theorem — Reading • 15 min
- Sum Rule — Reading • 4 min
- Product Rule (Chain Rule) — Reading • 2 min
- Bayes' Theorem — Reading • 43 min

Lesson 2: Gaussian Distribution

- Univariate Gaussian Distribution — Reading • 4 min
- Gaussian Distribution: Foundations and Applications — Reading • 15 min
- Multivariate Gaussian Distribution — Reading • 10 min
- Conditional and Marginal Multivariate Gaussian Distributions — Reading • 45 min
- **Product of Gaussian Densities** — Reading • 26 min

Lesson 3: Inference Techniques

- Bayesian Inference — Reading • 1h 10m
- Latent-Variable Models — Reading • 50 min
- Check Your Knowledge: Inference Techniques — Graded Assignment • 10 min

Module 4
Introduction to Model Evaluation and Optimization

Product of Gaussian Densities

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The product of two Gaussian (normal) distributions results in another Gaussian distribution. This property is particularly useful in various fields such as Bayesian statistics, signal processing, and machine learning, especially in cases like Kalman filters, Gaussian processes, and probabilistic inference.

General Formulation

Given two Gaussian distributions:

$$p1(x) = N(x \mid a, A)$$

$$p2(x) = N(x \mid b, B)$$

The product of these two distributions is also a Gaussian distribution, scaled by a constant:

$$p(x) = p1(x) \cdot p2(x) = c \cdot N(x \mid c, C)$$

Covariance Matrix C:

$$C = (A^{-1}) + B^{-1})^{-1}$$

This equation indicates that the precision (inverse of the covariance) of the resulting Gaussian is the sum of the precisions of the original Gaussians.

Mean c:

$$c = C \cdot (A^{-1a} + B^{-1b})$$

This shows that the mean of the resulting Gaussian is a weighted average of the means of the original Gaussians, where the weights are given by the precisions.

Scaling Constant c:

The scaling factor, which normalizes the resulting distribution, is given by:

$$c = (2\pi)^{-\frac{D}{2}} |A + B|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(a-b)^\top (A+B)^{-1}(a-b)\right)$$

In practice, it's often convenient to express the result of this product using the compact notation:

$$p(x) = N(x \mid c, C)$$

Even when x is not a random variable but rather a point or a parameter, this notation is used for simplicity and clarity.

Understanding the Product of Gaussian PDFs vs. Gaussian Variables

Read this discussion to explore the relationship between Gaussian probability density functions (PDFs) and Gaussian random variables. The post explains why the product of two Gaussian PDFs results in another Gaussian PDF, while the product of two Gaussian random variables does not necessarily yield a Gaussian distribution. This resource provides mathematical insights and practical examples to deepen your understanding.

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